

Part B-1

ATTUNE: Personalised anticipation of operator cognitive state for adaptive task decisions

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1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious and go beyond the state of the art)

People increasingly work with machines and computer systems that support decisions, coordinate tasks or act with some degree of autonomy. Even when parts of the work are automated, many situations still require human judgement, oversight or intervention, especially when conditions change or the consequences of a poor decision are serious. During time-critical work, the demands placed on the person can rise quickly and lead to slower decisions or errors. A human-in-the-loop system can therefore do more than personalize how information and requests are presented. A personal model can also guide the control layer when it decides how the work should proceed. Automation and other forms of system support can help, but support given at the wrong time or in the wrong form can interrupt work, increase attention switching or hide important information [10]. This challenge is also relevant to Industry 5.0, which places worker wellbeing, human control and resilience at the center of technology development [12].

Consider a team of operators supervising several robots during a time-critical operation. As the situation changes, one robot may report a fault, another may require manual control and a new task may arrive. This creates a difficult decision about who or what should handle each task. A task may be assigned to one of the operators, given to a robot with greater autonomy, or delayed until a better moment. More automation is not always better for the operator [10], so these options must be compared before a decision is made. A reactive approach changes the plan only after an operator is already overloaded. ATTUNE will instead predict how each available choice may affect each operator's workload and task performance, then select and apply the most suitable option.

Several obstacles prevent this. First, a person's cognitive state changes over time and cannot be observed directly. Second, the same task demands can affect people differently. Third, EEG can provide useful information about cognitive state, but it is not practical for routine use. Finally, the current cognitive state is only a starting point. It does not show how each possible action will affect the person over the next few minutes. This requires a personal model of how that person responds when task demands change over time.

Current systems mainly estimate the operator's present mental workload [1,2]. Many use one model learned from a group, although subject-specific models can improve workload estimation and eye responses can differ between people [3,9]. No single physiological measure is reliable in every setting [8]. These findings show that present cognitive state can be estimated, but current methods do not predict how each available action will change a particular person's state over the next few minutes. A controller needs this prediction to decide what to do now.

Previous work by Porghoveh et al. [11] developed a rolling-horizon task-allocation method that chooses actions by considering both their immediate effect and their consequences over a future time window. ATTUNE builds on this decision method by learning how each operator responds from real measurements and using that personal model to predict the effect of alternative actions before one is selected.

ATTUNE will first describe each available option through monitoring, attention switching, time pressure and decision difficulty. During calibration, EEG, eye tracking, task performance and brief self-reports will be used to learn how each person responds when these task demands change. During routine operation, only eye tracking and task data will be needed. ATTUNE will primarily use the learned personal response to predict the operator's workload and task performance under each available option. Stress and fatigue will be examined as secondary outcomes. The decision will therefore be based on the predicted effects on the person and task performance, rather than on the type of task or the application in which it is performed. This separation supports the use of the same decision method in different working environments.

Overall aim. ATTUNE aims to develop and validate a personal model that attunes to how each operator responds to changing task demands. Before any change is applied, the model will compare the operator's predicted workload and task performance under the available options. These predictions will support the selection of the most suitable option. Stress and fatigue will be evaluated as secondary outcomes. During routine operation, the method will rely only on eye tracking and task data.

Central hypothesis. A personalized nonlinear state-space model, calibrated with EEG, eye tracking, task performance and brief self-reports, can use eye tracking and task data alone during routine operation to predict how alternative task changes will affect an operator's workload over the next 60 seconds more accurately than non-personal, current-state and recurrent comparison models, while preserving task performance. The personal

dynamics learned during calibration are also expected to remain useful, without new calibration, in the second VR task.

Objective 1: Define and validate a common description of task demand. Each VR task will be described by monitoring, attention switching, time pressure and decision difficulty. The values will be calculated from task settings and logs using fixed rules. Each value will then be changed separately in pilot trials. Success will be evaluated by confirming that the planned change appears in the task logs and produces the expected change in participants' ratings relative to their own baseline. A task-demand value will be retained only if the logs and ratings show that it can be changed separately. If changing one demand also changes another in the same way, the two will be combined or the less useful one will be removed. Differences in how strongly people respond to the same demand will be learned by the personal model.

Objective 2: Build a personal model for routine use with eye tracking. During calibration, a multimodal teacher model will combine EEG, eye tracking, brief self-reports and task performance to estimate how each participant responds to the task demands. An eye-only student observer will then learn to reproduce the teacher's workload estimates using only eye tracking and task logs. The participant's personal response parameters will be carried into this routine model. Success will be evaluated by comparing the student model with a non-personal eye-tracking model that uses the same response values for everyone and with the multimodal teacher model. The student model must predict more accurately than the non-personal model. Its difference from the teacher model will show how much predictive accuracy is lost when EEG and self-reports are removed from routine operation.

Objective 3: Predict the effect of each available action before the control layer acts. At each decision point, ATTUNE will compare the operator's predicted workload and task performance if the control layer keeps the current task arrangement or applies one of the available actions. At predefined safe decision points, one feasible reversible action will be applied at random so that the predicted change can be compared with what actually happens after the action. The order of the no-change, reactive and ATTUNE conditions will be counterbalanced across participants. Predictions will be tested on data not used to build the model and compared with simpler forecasting methods and recurrent neural-network models trained with the same data. Success will require ATTUNE to predict the observed effects of the randomly applied actions more accurately than these comparison models. A change will be considered beneficial only if workload is reduced while task performance remains within the expected variation of the no-change condition.

Objective 4: Test whether the personal model works in a different VR environment without new calibration. The model learned in the VR robot-supervision experiment will be applied in a VR building-maintenance task, where participants monitor everyday maintenance requests and decide their order using simple information about urgency, deadlines, the building context and the number of people affected. Only eye tracking and task data will be used to make predictions, and the personal model will not be estimated again. Success will be evaluated by showing that the transferred personal model predicts workload more accurately than the non-personal model. Its prediction error will also be compared with the error obtained in the robot-supervision task to show how much accuracy is lost in the new task.

Workload will be the main outcome. A change will be considered beneficial only if it reduces workload while task performance remains within the expected variation of the condition in which no change is made. Stress and fatigue will be evaluated as secondary outcomes.

Ambition beyond the state of the art

- **Scientific novelty.** ATTUNE moves from estimating the current state to predicting how the available actions may affect a particular person. It also tests which parts of the model need a value learned for each person and which can use one shared group value.
- **Technological advance.** EEG will be needed only when the personal model is first built. After this stage, ATTUNE will use only eye tracking and task data. Before changing the task, the control layer will compare the available actions with keeping the task unchanged. If the predictions do not show which action is better, no change will be made. The operator will also be able to reject the proposed change.
- **Interdisciplinary impact.** Control engineering supplies the prediction and decision rule. Human factors defines the task demands and outcomes. HCI shapes the experiments and gives the operator control. Machine learning connects the full calibration data to a simpler model for routine use.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, the gender dimension and other diversity aspects, and open science)

The methodology is organized into five connected stages and two VR experiments. The VR robot-supervision experiment includes calibration, development of the personal model and the prediction test. A second, shorter VR experiment then tests whether the same personal model can be used without repeating calibration.

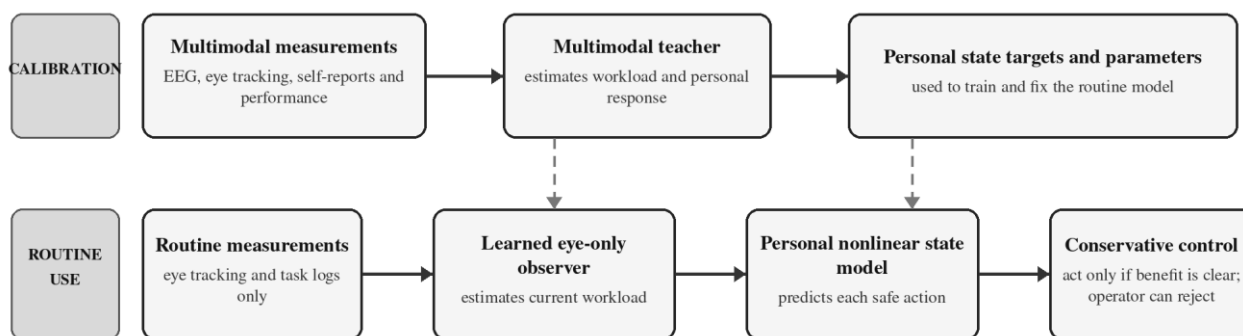


Figure 1. The multimodal teacher learns personal state targets and parameters during calibration. During routine use, the eye-only observer and personal nonlinear state model predict the effect of each safe action before conservative control.

Stage 1: Define and validate the task demand values

Task settings will be converted into candidate values for monitoring, attention switching, time pressure and decision difficulty. Monitoring will reflect how many items require attention. Attention switching will reflect the required shifts between tasks or displays. Time pressure will compare the time available with the expected processing time of each task. Decision difficulty will reflect how many choices must be compared, how similar they are, how likely each choice is to succeed and the possible consequences of choosing incorrectly [4,5]. Each candidate task demand will be tested in pilot trials. To test monitoring, for example, the number of items requiring attention will be increased, while the required attention switches, available response time and decision choices are kept as similar as possible. The task logs will record what actually occurred in each trial. After each trial, participants will rate how much monitoring, attention switching, time pressure and decision difficulty they experienced. For each participant, the rating from the higher-demand trial will be compared with their rating from the matched lower-demand trial. A task-demand value will be retained only if the logs and ratings show that it can be changed separately. If changing one demand also changes another in the same way, the two will be combined or the less useful one will be removed. The pilot trials will also be used to choose the number and length of task runs for the main experiment. The experiment must include enough changes in task demand to estimate the personal model reliably, without making each visit too long or tiring for participants. The final design will be fixed before the main experiment is preregistered.

Stage 2: Run the VR robot-supervision experiment and collect calibration data

Approximately 25 healthy adults will be recruited for the VR robot-supervision experiment. The exact target will be fixed before the main experiment, using the pilot data to estimate how many participants are needed for reliable model comparisons. During calibration, EEG, eye tracking and task performance will be recorded while the task demands are changed in a random and balanced order. At fixed points, participants will separately rate their current workload, stress and mental fatigue on short scales from 0, meaning none, to 10, meaning very high. These ratings will give the personal model reference values for the participant's state at those points in the task. After each task run, NASA-TLX will be used to assess the workload experienced across the complete run [6]. Each participant will repeat a short calibration on another day. The repeat will test whether the learned response belongs to the person rather than to that particular day. Both calibration visits are part of the same VR experiment. The calibration data will be accepted when the recordings are complete enough for analysis and each segment of EEG and eye-tracking data can be linked to what was happening in the VR task at that time, such as an alert appearing or the participant making a choice. The task logs must also show that each planned change actually occurred. For example, in a trial designed to increase monitoring demand, the logs must show that the participant was given the planned additional items to monitor.

Stage 3: Build and test the personal model

ATTUNE will be formulated as a personalized nonlinear state-space model. Its state equation, $x_i(t+1) = f(x_i(t), d(a_t); \theta_i)$, will describe how participant i 's hidden workload state changes when action a_t changes the task demands d . The nonlinear function will allow the effect of several demands together to differ from the sum of their separate effects and will keep predictions within the valid workload range. Workload will be the primary state used by the control layer; stress and fatigue will be estimated as secondary outcomes. The observation model will connect the hidden state to EEG, eye tracking, brief self-reports and task performance. During calibration, a multimodal teacher will use all these measurements to estimate the participant's changing state and personal response parameters θ_i . An eye-only student observer will then learn to reproduce the teacher's workload estimates from eye tracking and task logs alone. During routine use, the learned observer and personal state equation will be used without EEG or self-reports. The model will initially test 21 parameters that may differ between people: 12 gains describing how strongly the task demands affect workload, stress and fatigue; three time constants describing how quickly these states change; three eye-tracking baseline values that account for each person's usual eye response; and three output offsets that adjust the predicted levels of workload, stress and fatigue. The 12 gains will be checked for a shared pattern. For example, some participants may respond more strongly to all task demands, rather than requiring a completely different response value for every combination of task demand and cognitive state. If this simpler pattern explains the data equally well, the model will use fewer personal parameters. For each model parameter, ATTUNE will test whether a different value is needed for each participant or whether one common value is enough for everyone. A different value will be used only if the model can estimate it correctly from simulated data, gives a similar result when the same participant repeats the calibration on another day, and improves predictions for that participant's separate test data. To test one participant fairly, data from all the other participants will first be used to build the general model and estimate the common parameter values. The selected participant's calibration runs will then be used to estimate their personal parameter values. Their other task runs will be kept separate and used to compare two predictions: one using their personal values and one using the common group values. If the personal values produce more accurate predictions, they provide evidence that personalization is useful. The process will be repeated until every participant has been tested once. If a parameter fails any of these tests, ATTUNE will not estimate a separate value for each person. The parameter will remain in the model, but the same group value will be used for everyone. Success will be evaluated by comparing the personalized routine model with two alternatives. The first is a non-personal eye-tracking model that uses the same response values for everyone. The second is the calibration model, which uses EEG, eye tracking, self-reports and task performance. The personalized routine model must predict more accurately than the non-personal model. Its difference from the calibration model will be reported to show how much predictive accuracy is lost when EEG and self-reports are removed from routine operation.

Stage 4: Test action-effect predictions and compare decision-making principles

The prediction-test visit will contain two connected parts. In the first part, ATTUNE will predict the effect of each feasible reversible action at predefined safe decision points. One action will then be selected at random and applied by the control layer. The observed changes in workload and task performance will be compared with the changes predicted before the action was selected. Random selection prevents the test from including only the actions that the controller already expects to be best. ATTUNE's predictions will also be compared with a model that assumes workload remains at its current level, a model using only current eye features, and LSTM and GRU forecasting models trained with the same data. All models will be tested on the same data that were not used to build them.

In the second part, the experiment will compare three decision-making principles. The no-change principle keeps the current task arrangement. Under the reactive principle, the control layer selects an action only after high workload has been detected. Under the ATTUNE principle, the control layer predicts how each available action may affect that participant's workload and task performance before selecting an action. The available actions depend on the application and may include delaying, reallocating, simplifying or dividing a task, changing the level of automation, or leaving the task unchanged. The order of the three conditions will be counterbalanced across participants so that practice, fatigue or presentation order does not favor one condition. Both parts belong to the same prediction-test visit within the VR robot-supervision experiment. At each decision point, ATTUNE will estimate the operator's current state from recent data and predict action effects over the next 60 seconds. If the operator accepts the selected action, the control layer will apply it and repeat the prediction when new data arrive. Predictions at 30 and 120 seconds will test whether the results depend on the selected horizon, while fatigue will be evaluated across the complete task runs.

Stage 5: Short transfer test without new calibration

Participants will complete a brief VR building-maintenance prioritization task. They will monitor everyday maintenance requests, such as a water leak, heating failure, broken appliance or routine inspection. They will decide which request should be handled first and assign the available maintenance support using simple information about urgency, deadlines, the building context and the number of people affected. No technical diagnosis or repair knowledge will be required. The personal parameter values learned for each participant during the robot-supervision calibration will be transferred unchanged to the building-maintenance task. No parameter values will be recalculated using data from the second task. No EEG calibration or model development will be repeated. Before the transfer test, a short pilot will check whether the new VR task produces the intended task demands. Task logs will calculate the demand values, and participants will briefly rate the monitoring, attention switching, time pressure and decision difficulty they experienced. These ratings will be used only to check the mapping between the new task and the common demand description. They will not be used to change the personal model. During the transfer test, ATTUNE will still make its predictions using only eye tracking and task data. State ratings and task performance will be collected only to evaluate the predictions. Because this stage reuses the participants, personal model and task-demand description from the main experiment, it will require considerably less effort. Transfer will be evaluated against the non-personal model, and its prediction error will be compared with the error obtained in the robot-supervision task. This will show both whether the personal model remains useful and how much accuracy is lost in the new task.

Analysis integrity and safe evaluation. Calibration and test runs will be kept separate. To check personalization, each participant's test data will be predicted using two versions of the same model. The personalized version will use the values estimated from that participant's calibration data. The non-personal version will use one shared set of values estimated from the other participants. Personalization will be considered useful only if the personalized version gives more accurate predictions. Prediction errors from held-out calibration runs will be used to set an uncertainty limit before the prediction test. The control layer will apply a change only when the predicted reduction in workload is larger than this limit and task performance is protected. Otherwise, it will keep the current task arrangement or ask the operator for confirmation. All tests will use controlled VR tasks with no safety risk. The final analysis will report both the size of the differences between the models and the uncertainty around those differences.

Interdisciplinary integration. Human factors and HCI define and validate the task demands. Experimental HCI and physiological sensing provide the calibration data. Control engineering defines how the system state changes over time and how the control layer chooses an action. Machine learning estimates the model parameters and compares the forecasting methods. These contributions form one connected method.

Gender dimension and other diversity aspects. ATTUNE will not assume that one eye tracking model performs equally for everybody. Recruitment will consider sex, gender and age where feasible. Corrective eyewear will be recorded with consent. Signal quality, missing data and model error will be checked across these factors. Results will be separated by group only when the sample supports a meaningful interpretation.

Open science, ethics and data governance. Protocols, hypotheses, exclusions and analysis plans will be preregistered. Code, demand manuals, model cards and suitable derived or synthetic data will be released when consent and reidentification risk allow, following FAIR principles [7]. Physiological and behavioral data will be pseudonymized and stored on controlled AAU systems. ATTUNE estimates task related workload and fatigue. It does not infer emotion or support employment decisions. Participants will see when a task change is active and can pause or reject it.

1.3 Quality of the supervision, training and of the two way transfer of knowledge between the researcher and the host

Supervision and scientific fit. Associate Professor Hendrik Knoche provides the expertise ATTUNE needs in human-computer interaction, human-robot interaction, participatory design, eye tracking, VR, BCI and experimental evaluation. His expertise complements my background in control, optimization and modeling. The Human Machine Interaction group and Human Robot Interaction laboratory provide the setting needed to turn my control method into a scientifically valid study with people. I am already a research visitor in CREATE, so the working relationship and local integration have begun. AAU has approved my affiliation on the MSCA platform; ethics and scientific approvals will follow normal AAU procedures.

Supervision and career mentoring. Knoche and I will meet every two weeks to review scientific decisions and current results. A monthly review will track tasks, data quality, decision gates and risks. Career progress will be reviewed every three months, supported by the Individual Career Development Plan agreed in M1 and formally reviewed in M6, M12, M18 and M24. I will also take part in group seminars and laboratory activities.

Training from the host to me. The training directly addresses the main gap in my profile: designing and evaluating human-centered systems.

- **Human centered research:** task analysis, participatory design, experimental HCI, counterbalancing and construct validation.
- **Multimodal experiments:** eye tracking and EEG preprocessing, synchronization, signal quality checks and uncertainty calibration.
- **Responsible research:** ethics for studies with people, GDPR, FAIR data, bias checks and transparent task changes.
- **Career development:** project management, supervision, teaching, grant writing, science communication and open source stewardship.

At least 40 hours of formal training will be documented. Each activity will produce evidence of use in the project, such as a validated protocol, an analysis plan, a teaching activity or a grant draft.

Knowledge from me to the host. I will bring optimization, rolling-horizon task assignment, control, signal processing, BCI prototyping and reproducible numerical modeling. I will share these skills through an internal tutorial in M2, reusable simulator components, quarterly code and design clinics, supervision of one MSc project and mentoring of one BSc project.

How knowledge will be exchanged. The host will train me in HCI study design, participatory methods, eye tracking, research governance and responsible deployment from M2 to M3 and through continued study reviews. A joint workshop in M12 and one guest lecture will connect the two research traditions. Tutorial material, simulator components and analysis guidance will remain in the group's repositories so the exchanged knowledge remains available after the fellowship.

No secondment or non academic placement is proposed. The focused 24 month plan already contains the complementary expertise and facilities needed at AAU.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Dual doctoral training. I hold a PhD in Mechanical Engineering from Shahid Chamran University of Ahvaz and am completing a second PhD in Autonomous Systems at Politecnico di Bari. My first PhD gave me depth in mechanical modeling, vibration and acoustics, robotics, control and optimization. My second PhD expanded this foundation toward autonomous decision-making and the interaction of people with robots and computer-based systems. This combination allows me to study ATTUNE at two connected levels: how the control layer makes and carries out decisions, and how those decisions affect the person working with the system.

International and interdisciplinary trajectory. My research journey has taken me from Iran to Italy, Türkiye and Denmark. During these periods, I have worked with researchers in mechanical engineering, control, robotics, human-computer interaction, biology and chemical engineering. Working across institutions and fields has taught me to adapt to different research cultures, explain my methods to researchers outside my field and combine different forms of expertise around a common problem. This experience aligns with the MSCA emphasis on international mobility, interdisciplinary research and two-way knowledge exchange.

Research results. My research has produced results beyond conventional publications. In robotics, I contributed to Iranian patent IR 95162 for a controllable humanoid warning and tracking robot. In an environmental research project, I worked with biologists and chemical engineers to develop suitable experimental flow conditions for a Spirogyra biofilter. This collaboration contributed to the registration of a partial small-subunit ribosomal RNA sequence in the NCBI GenBank database under accession MW063644 [13]. The patent shows my ability to turn an engineering concept into a working system. The NCBI record demonstrates that I can make a productive technical contribution outside my original discipline.

Publications and scientific visibility. My publications cover nonlinear vibration, active noise control, acoustic reconstruction, robotics, optimization, autonomous systems, brain-computer interfaces and human-robot interaction. My recent work includes rolling-horizon task allocation for human-robot teams [11], a compact P300 interface for mission-level robot control and a review of optimization and control in robotized logistics. I have published in peer-reviewed journals and presented my research at international IEEE conferences. This record shows a consistent progression from modeling physical systems to designing autonomous systems that consider the people working with them.

Open science. I maintain two public code repositories. The Acoustic Enclosure Control repository contains material related to my acoustic modeling and control research. The Cognitive-State-Aware Task Allocation repository provides the base implementation of my rolling-horizon task-allocation research. These repositories will be further developed with clearer documentation, reproducible examples and suitable sample or synthetic data. ATTUNE will build on this practice by releasing documented code, task-demand rules and suitable derived or synthetic data when ethical and legal requirements allow.

Technical skills. I have experience obtaining useful results from limited and noisy experimental data. My methods include signal denoising, wavelet analysis, modal analysis, parameter estimation, feature extraction, feature selection and reducing the number of variables used by a model. I use Python, MATLAB, COMSOL and Fortran for modeling, simulation, data analysis and experimental validation. These skills are directly relevant to ATTUNE

because each participant will provide only a limited amount of calibration data. The model must use these data to determine which response values genuinely need to be personalized.

Leadership and research independence. As first author on several studies, I have taken responsibility for mathematical formulation, programming, analysis, validation and manuscript preparation while coordinating contributions from collaborators. At Politecnico di Bari, I have connected control, logistics, BCI and human-robot interaction within productive collaborations. This experience shows that I can lead my research independently while also contributing constructively to an interdisciplinary team.

Scientific service, mentoring and teaching. I have reviewed manuscripts for Biomedical Signal Processing and Control. At Shahid Chamran University and Politecnico di Bari, I have taught technical subjects and supported students in modeling, optimization and programming. These activities have developed my ability to explain complex methods, guide technical work and contribute to a productive research environment. ATTUNE will build on these strengths while providing the systematic training in HCI study design, participatory methods and responsible human-centered control that I still need.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his skills development

This fellowship will strengthen my career prospects by combining advanced scientific training, structured mentoring and clear steps toward research independence. It will prepare me for academic careers in human-robot interaction, human-computer interaction and autonomous systems, as well as industrial research roles in robotics, decision support and human-centered AI.

In M1, I will prepare an Individual Career Development Plan with my supervisor. The plan will set measurable goals for research training, publications, teaching, student supervision, networking and future funding. We will review the plan in M6, M12, M18 and M24. At each review, we will record which goals have been completed, which are delayed and what action is needed next.

By the end of the fellowship, I will have led a preregistered VR robot-supervision experiment and a shorter transfer test in a different VR task. I will also have developed and evaluated a personal workload-prediction model, compared it with non-personal and learning-based alternatives, and tested how its predictions can guide decisions made by the control layer. From M18 to M24, I will prepare an independent funding application and a research statement on human-centered autonomous systems. These outputs will provide clear evidence of my ability to lead interdisciplinary research independently.

The planned career-development steps and the capability built by each step are summarized in Table 1.

Table 1. Career-development measures and expected gains.

Period	Career measure	Expected gain
M1 to M6	Baseline skills review; HCI and ethics training; first supervision task; personal publication and networking plan	A clear development gap and measurable plan
M7 to M15	Lead the VR robot-supervision experiment; present methods internally and at one international venue; mentor a student on the simulator	Evidence of research leadership and teaching
M16 to M24	Lead the prediction test and the short building-maintenance transfer test; organize the stakeholder workshop; prepare an independent grant and research statement	Competitive academic or industrial research profile

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, including dissemination, exploitation and communication

The plan follows the project stages. Early activities will make the protocol and task-demand description visible. Later activities will communicate the validated model, software and evidence. Table 2 links each route to its audience and timing.

Scientific dissemination and open outputs. Three open-access manuscripts will report the validated demand description and calibration study, the personal dynamic model, and the prediction and transfer results. Results will also be presented at suitable HCI, HRI, control and human-factors venues. Preregistrations, the demand manual, simulator, model code, model card and suitable derived or synthetic data will support verification and reuse.

Exploitation and protection. The main reusable results will be an open reference implementation and an integration guide for researchers and developers. Before release, AAU innovation support will check whether any result requires protection. No patent or company partner is assumed; if a protectable result appears, it will be assessed before public disclosure.

Communication. A project page and three short visual explainers will present the problem, method and limits in plain language. An AAU public-science activity and a stakeholder workshop will use a controlled demonstration to show what the system changes, what information it uses and how the operator can reject a change.

Table 2. Dissemination, exploitation and communication plan.

Route	Measure	Audience	Timing
Dissemination	Three open-access manuscripts and presentations at suitable HCI, HRI, control and human-factors venues.	Researchers	M8-M24
Open outputs	Preregistrations, demand manual, simulator, documented code, model card and suitable derived or synthetic data.	Researchers and developers	M3-M24
Exploitation	Open reference implementation and integration guide, following an AAU review of any protectable result.	Adaptive-system developers	M16-M24
Communication	Project page, three short explainers, an AAU public-science activity and a stakeholder workshop with a live demonstration.	Public, students, operators and stakeholders	M2-M24

Monitoring. Progress will be checked against three submitted manuscripts, reusable open outputs, at least 40 stakeholder-workshop participants and about 100 public participants or online viewers. At the M20 workshop, independent users will apply the demand manual to unseen task configurations. Their disagreements and requested clarifications will show where the manual still needs improvement.

Responsibilities. I will lead the manuscripts, repositories, tutorials and public materials. Knoche will provide scientific quality control and access to HCI and human-robot interaction networks. AAU services will support data protection, communication and any intellectual-property review before release.

Responsible communication. Every public message will state that ATTUNE predicts task-related workload under controlled conditions. It does not read emotions or judge workers. Consent, data protection and the limits of the evidence will take priority over visibility.

2.3 Magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

ATTUNE's expected impact is limited to what can be demonstrated within 24 months. The project will deliver evidence, reusable methods and a controlled prototype, not a claim of proven workplace effectiveness. Table 3 connects each contribution to evidence produced during the fellowship.

Scientific impact. Current adaptive systems mainly estimate the operator's present state. ATTUNE will add a controlled test of whether the effect of a possible controller action can be predicted before that action is applied. The validated demand description, randomized action-effect data and model comparisons will give other researchers a common basis for studying temporal prediction and personalization.

Technological and economic impact. EEG is used to build the richer calibration model, while routine prediction uses eye tracking and task data. The teacher-to-student training and conservative control rule will show whether useful decisions can be supported with fewer sensors and without acting when the predicted benefit is uncertain. The transfer test will also measure how much accuracy is lost when the same personal model is reused in another VR task without new calibration.

Societal impact. The operator can see and reject task changes, and the system is not designed to infer emotion or support employment decisions. Reporting errors across participant groups and publishing the model's limits will provide a more responsible basis for any later field study.

Table 3. Expected impacts, pathways and evidence within the fellowship.

Impact	Contribution and pathway	Evidence within the fellowship
Scientific	A common demand description, causal action-effect data and a personal temporal model support comparison of static, temporal and dynamic approaches.	Validated manual, model comparisons, open benchmark and peer-reviewed outputs.
Technological and economic	An eye-only routine model and conservative control reduce the sensing burden; transfer without recalibration tests reuse in another VR task.	Eye-only prediction error, controller comparison and transfer error.
Societal	Visible, rejectable task changes and explicit limits provide a human-centered route to assistance without emotion inference or employment scoring.	Override records, model-error checks across participant groups and documented limits.

The project will therefore not claim workplace effectiveness from a laboratory study. Its 24-month contribution is a validated method, evidence about where forecasting works or fails and a controlled prototype. These results will provide a credible basis for deciding whether a later field trial is justified.

3. Quality and efficiency of the implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Work-plan logic and critical path. WP1 fixes the task-demand description before the main data collection. WP2 collects the calibration data and builds the richer calibration model. WP3 decides which model parameters genuinely require personal values and fixes the personalized routine model before the prediction test. WP4 then tests the forecasts and applies the frozen personal model in the shorter transfer experiment. WP5 runs throughout the fellowship. WP2 may begin simulator preparation before WP1 ends, but main data collection cannot begin until the demand rules and protocol have passed the WP1 gate.

Effort allocation. The total effort is 24 person-months. WP2 to WP4 receive most of the effort because they contain participant visits, signal processing, model identification and controlled prediction tests. WP1 receives four person-months for pilot work, protocol design, ethics and preregistration. WP5 receives three person-months distributed across the fellowship for management, training, data governance, dissemination and career development.

Monitoring and responsibility. I will lead the daily work and maintain the task, decision-gate and risk records. Knoche and I will review scientific progress every two weeks and formally check outputs, gates and risks each month. Career development and project scope will be reviewed every three months. A failed gate will activate the stated fallback and narrow the claim; the next stage will not continue as though the gate had been passed.

Table 4 translates the work-plan logic into activities, outputs, decision gates and fallbacks.

Table 4. Work packages, outputs, decision gates and fallbacks.

WP	Timing	PM	Activities and main output	Decision gate and fallback
WP1	M1-M5	4	Task-demand rules, pilot trials, protocol, ethics and preregistration Output: D1: validated demand manual and protocol	Each retained demand must be changed separately. Otherwise combine or remove it.
WP2	M4-M10	5	VR robot-supervision experiment, repeat calibration and full calibration model Output: D2: linked calibration dataset and full model	Signals and logs must be usable and linked to task events. Otherwise simplify secondary measures and replace unusable data within the approved target.
WP3	M8-M16	6	Parameter tests and personalized eye-tracking model Output: D3: benchmark and fixed personal/shared parameters	The routine model must beat the non-personal model. Otherwise retain the full calibration result and withdraw the routine-use claim.
WP4	M14-M22	6	Randomized action-effect test, counterbalanced decision-principle comparison and short transfer test Output: D4: controlled prototype and evaluation report	Predicted action effects must beat comparison models. If not, shorten the reliable horizon or remain in recommendation mode. Weak transfer restricts the claim to robot supervision.
WP5	M1-M24	3	Management, data governance, training, impact and career development Output: DMP, career reviews, communication and code release	Monthly and quarterly reviews trigger re-sequencing or scope reduction when a delay threatens a scientific gate.

The timing and overlap of the work packages are shown in Table 5.

Table 5. Twenty-four-month work-plan schedule.

WP	1 to 2	3 to 4	5 to 6	7 to 8	9 to 10	11 to 12	13 to 14	15 to 16	17 to 18	19 to 20	21 to 22	23 to 24
WP1	•	•	•	•	•							
WP2		•	•	•	•	•						
WP3				•	•	•	•	•				
WP4							•	•	•	•	•	
WP5	•	•	•	•	•	•	•	•	•	•	•	•

The main implementation risks, their early warnings and the planned responses are summarized in Table 6.

Table 6. Main implementation risks, early warnings and fallbacks.

Risk	L/I	Early warning	Mitigation and fallback
Demand values are not separable	M/H	Two ratings repeatedly change together or logs do not show the intended contrast	Revise the pilot rules; combine overlapping demands or remove the less useful one.
EEG or eye data are poor, or participant burden is high	M/M	Missing or noisy segments increase, or participants cannot complete planned runs	Synchronization uses the host laboratory's Lab Streaming Layer setup. Check quality during collection, shorten blocks and add breaks. Reduce secondary measures before reducing the primary workload test.
The eye-tracking model is too weak	M/H	It does not improve on the non-personal model in held-out data	Use a longer calibration window and structured features. Keep the richer calibration model and withdraw the routine-use claim.
Personal parameters are unstable	M/M	A value cannot be recovered or changes strongly on the repeat day	Keep the parameter in the model but use one shared group value.

Risk	L/I	Early warning	Mitigation and fallback
Participant loss or unusable recordings reduce the sample	M/M	Usable data fall below the preregistered target	Recruit replacements within the approved range and monitor data during collection. Protect the primary workload comparison and treat secondary outcomes as exploratory.
Forecasts over longer periods or transfer are weak	M/M	No gain over comparison models, or transfer error rises substantially	Shorten the supported horizon and keep ATTUNE in recommendation mode. If transfer is weak, claim use only in robot supervision.
Ethics approval or recruitment is delayed	L/H	Approval or recruitment is behind the monthly plan	Start review in M1, use established AAU channels and advance simulator, code and pilot preparation while participant work is paused.

3.2 Quality and capacity of the host institution, including hosting arrangements

Hosting arrangements and integration. I will receive workspace, computing access, software and access to the VR and sensing infrastructure from the start of the fellowship. My current research visit to CREATE means that the working relationship, local procedures and scientific integration have already begun. During the fellowship, integration will continue through HMI group meetings, seminars, shared experimental work and student supervision. The scientific review every two weeks and the monthly review of outputs, data quality and risks described in Section 1.3 will connect the hosting arrangements directly to project delivery.

Project-specific scientific capacity. AAU provides the combination required by ATTUNE: human-centered interaction research, eye tracking, human-robot interaction facilities and experience with EEG, BCI, VR and participant studies. Calibration recordings will use the Brain, Behavior, Architecture Research laboratory of Assoc. Prof. Zakaria Djebbara, which holds Waveguard EEG caps with ANT Neuro eego Sports amplifiers, Meta Quest Pro headsets with integrated eye tracking, and Lab Streaming Layer synchronization across all equipment. Synchronized multimodal recording is therefore already established rather than built for this project. The laboratory has already completed a related Whack-a-Mole study with 25 adults, providing practical evidence that a participant study of similar scale can be completed with the available facilities and support.

Institutional and administrative capacity. AAU has established procedures for research ethics, data protection, data management, open science, innovation review and EU project administration. AAU Research Services will support grant administration and reporting. The relevant institutional units will support consent and data-protection procedures, secure handling of participant data, open-output decisions and review of any potentially protectable result before release. AAU's approval of my host affiliation on the MSCA submission platform confirms the administrative hosting arrangement; scientific and ethics approvals will follow the normal AAU procedures.

Completeness of the hosting environment. No external secondment is required because the complementary expertise, participant-study facilities, computing support and institutional services needed for ATTUNE are available at AAU. This keeps responsibility for the scientific work, participant protection and project management within one coordinated host environment. Further institutional details and relevant infrastructure are provided in Part B-2.

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